The Jing-Zhang Gauge

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Urban Design Proposal for the Centennial Jing-Zhang AI Innovation Belt · A3 Booklet

A public specification as the core product

VERSION LINE · the full journey of one standard part

All areas are provisional recalculated values

GW
Gauge Works
Spec drafting · Conformance testing · Certification
192.1 ha

→

PB
Proto Blocks
First installation · Resident jury · Iterative rollback
104.3 ha

→

MF
Market Floor
Scaled deployment of certified parts
72.0 ha

Retest data flows back · triggers the next version

- **Access Wing · Zhongguancun Technology Service Wing** —— factors, IP and capital enter through the certification interface
- **Field-Test Wing · Xiaoyuehe Scenario Empowerment Wing** —— real-environment retests; failure cases are published

Key numbers · all recalculated from this package's GeoJSON and matrices, never copied from narrative

11.413 km² design scope (recalc · provisional)	**167** land-use parcels	**552** building footprints	**9** GeoJSON layers
16 AI scenario cards	**4** datum points (landmarks)	**7** global case studies	**8** deliverable modalities

READING MAP

boundary state: provisional

Suggested path: 02–03 for the big picture → 05 and 09 for key areas and scenarios → 07 to trace every number to its source. All drawings share one source with the text, GeoJSON and metrics.json.

01

Overall Concept and Naming System

agent.1 Overall concept and functional coordination

The concept

What this belt ultimately produces is not buildings but a GAUGE — a public specification for AI in urban space that anyone can build to, connect to, and audit against.

A century ago the Jing-Zhang Railway set its own technical specification without external control. Self-reliance -> standard-setting -> interoperability is its real technical legacy, and the two ends of the taskbook functions of full-stack autonomy and governance discourse power.

Naming system

The Jing-Zhang Gauge	overall concept
JZ-Gauge	versioned public specification set
JZ-Parts	catalogue of physical standard parts
JZ-Specs	scenario behavioural boundaries
Datum	honour and deliberation system
Gauge Week	annual specification release

Why gauge

A gauge produces no locomotive, yet decides which vehicles may run on the line. Whoever holds the standard gains authority over the network without owning a single vehicle.

Governance discourse power comes not from declarations but from others having to build to your specification.

Spatial structure

One line: the Version Line, a release pipeline made physical
Three stations: Gauge Works / Proto Blocks / Market Floor
Two interfaces: Access Wing (factors) / Field Wing (stress test)

North-south is not a traffic connection but a process connection.

Fig. 1 Overall Design Area: One Line, Three Stations, Two Interfaces

The Version Line links the three stations; six stitch points bridge the railway severance

Fig. 1 Overall Design Area: One Line, Three Stations, Two Interfaces

The Version Line links the Gauge Works, Proto Blocks and Market Floor stations; six stitch points bridge the severance across the railway

Version Line · Stitch point · Datum point · Key areas (provisional) · Design area (provisional) · Road network (along parcel edges)

Sources: brief/site-package (planning_limits.json, provisional_boundaries.geojson, agent_taskbook.json)

Status: provisional boundary; conceptual proposal, not a statutory output or official red line

THE JING-ZHANG GAUGE · JZ-Gauge v0.1 · EPSG:4548

Fig. 2 Land-Use Structure and Three-Level Scope Framework

Voronoi partition guarantees exact shared boundaries, full coverage, no gaps, no overlaps

Fig. 2 Land-Use Structure and Three-Level Scope Framework

Land use is generated by Voronoi partition; adjacent parcels share exact boundary coordinates. Residual gap 14.897 m², overlap 0.099 m²; not a complete partition

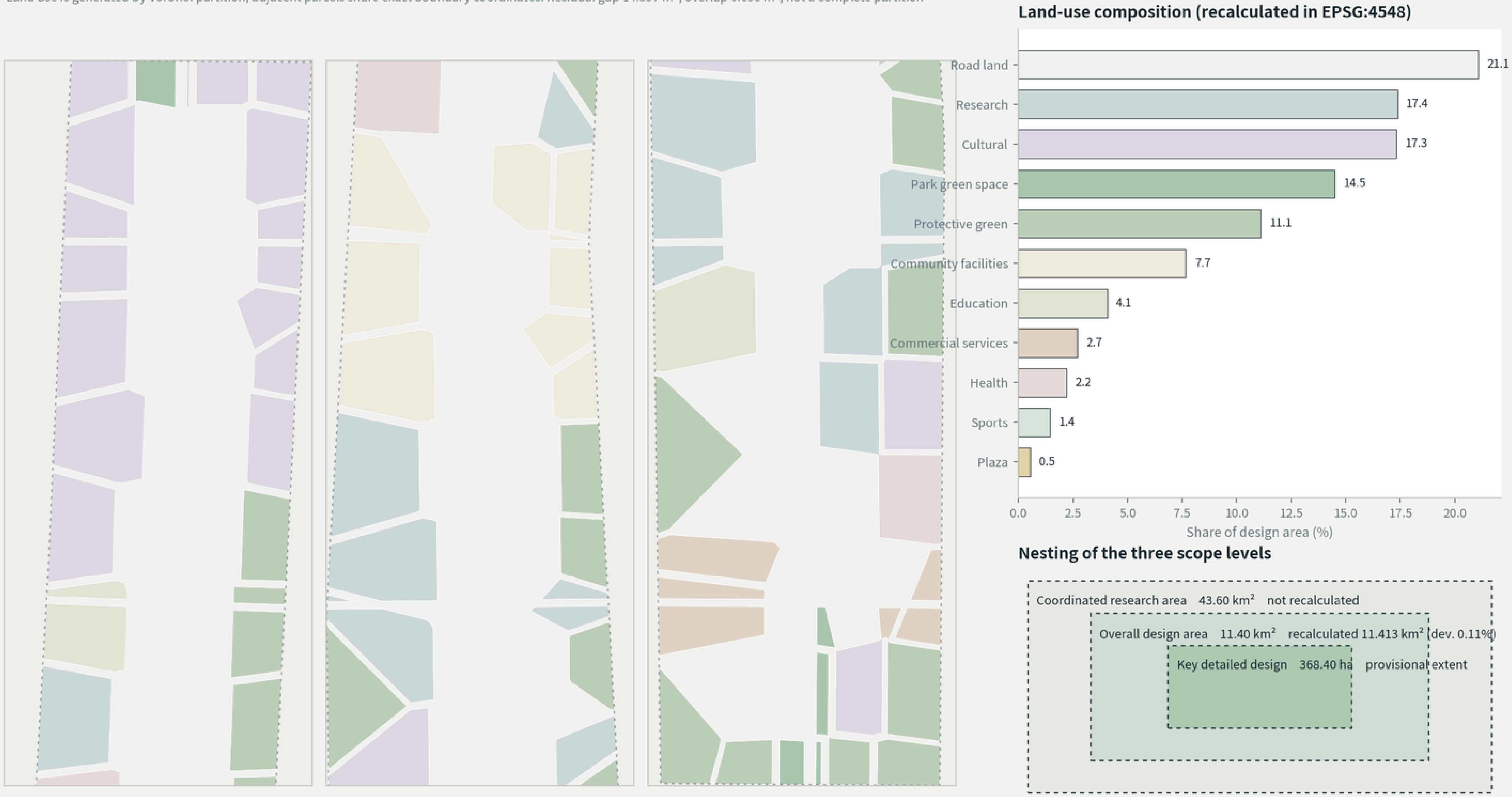

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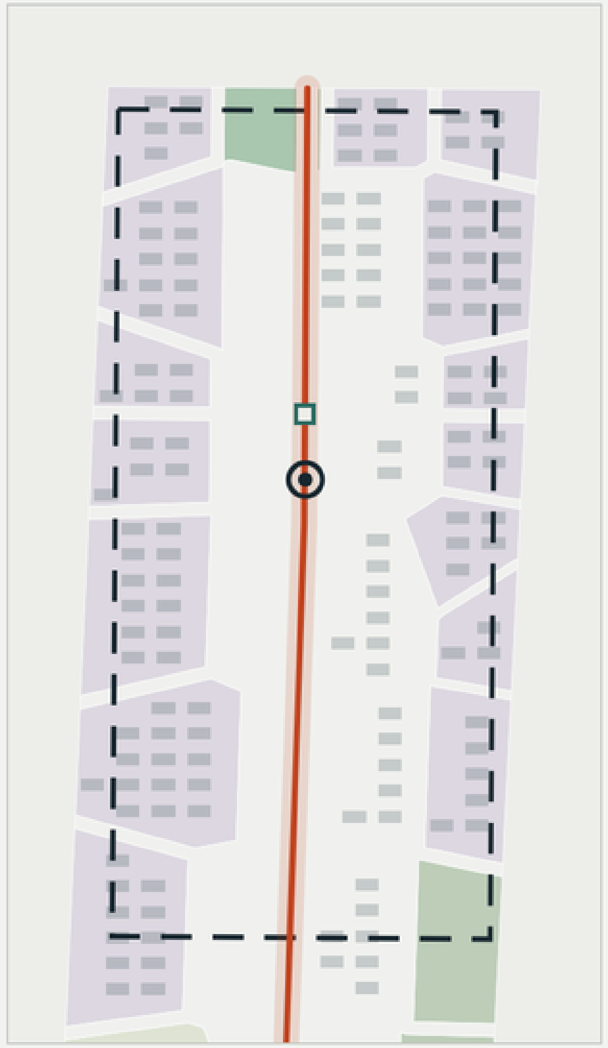
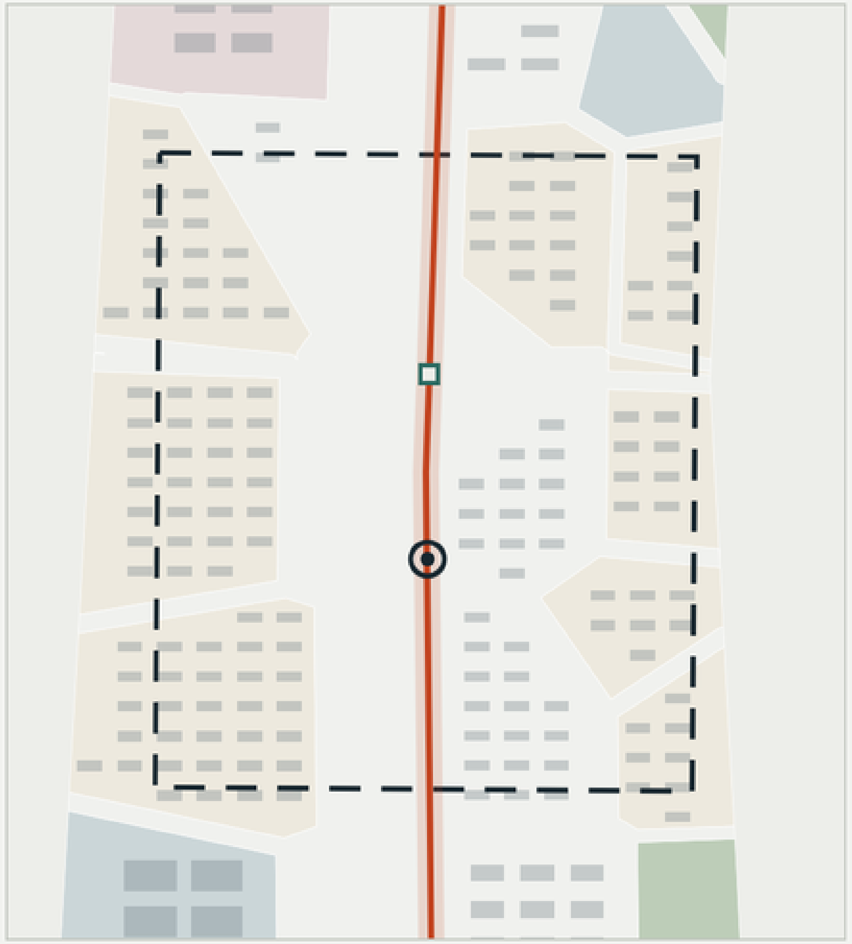
Fig. 3 Index and Design Tasks of the Three Key Areas

The three key areas are not parallel industrial blocks but three stations on one production line

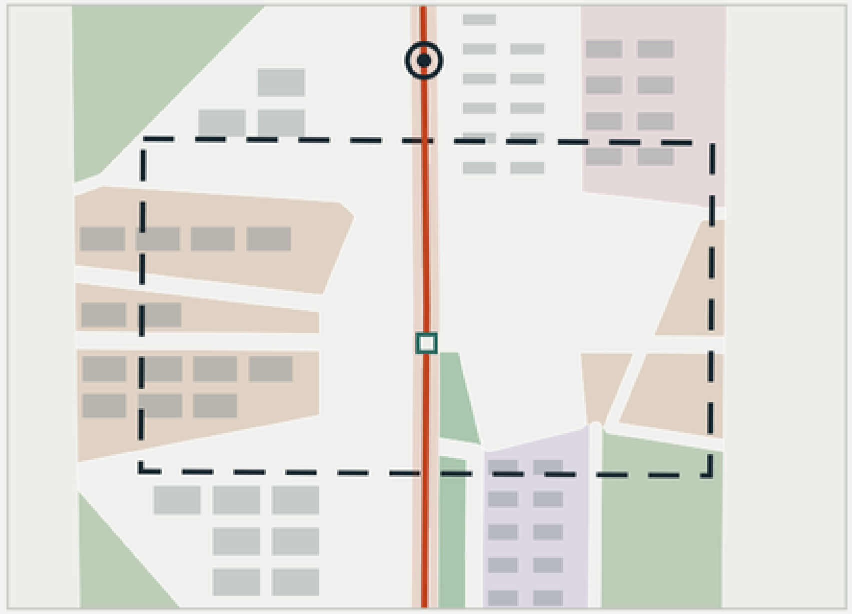
GW Gauge Works



PB Proto Blocks



MF Market Floor



Area (provisional): 192.1 ha
Core task: Specification drafting · Conformance testing · Red-team certification
Signature node: D-01 Datum Zero
Scenarios hosted: S-10 / T-01 / T-03

Area (provisional): 104.3 ha
Core task: First installation · Resident jury · Iteration and rollback
Signature node: D-02 The Objection Stand
Scenarios hosted: S-05 / S-09 / T-02

Area (provisional): 72.0 ha
Core task: Scale deployment of certified parts
Signature node: D-03 The Decommission Yard
Scenarios hosted: S-07 / S-12

Implementation risk: all three key-area polygons are provisional, so the conclusions above are directional design only; land-use boundaries, development scale and implementation projects must be re-established once official polygons are released.
This drawing provides no demolition conclusion, ownership judgement or engineering feasibility conclusion.

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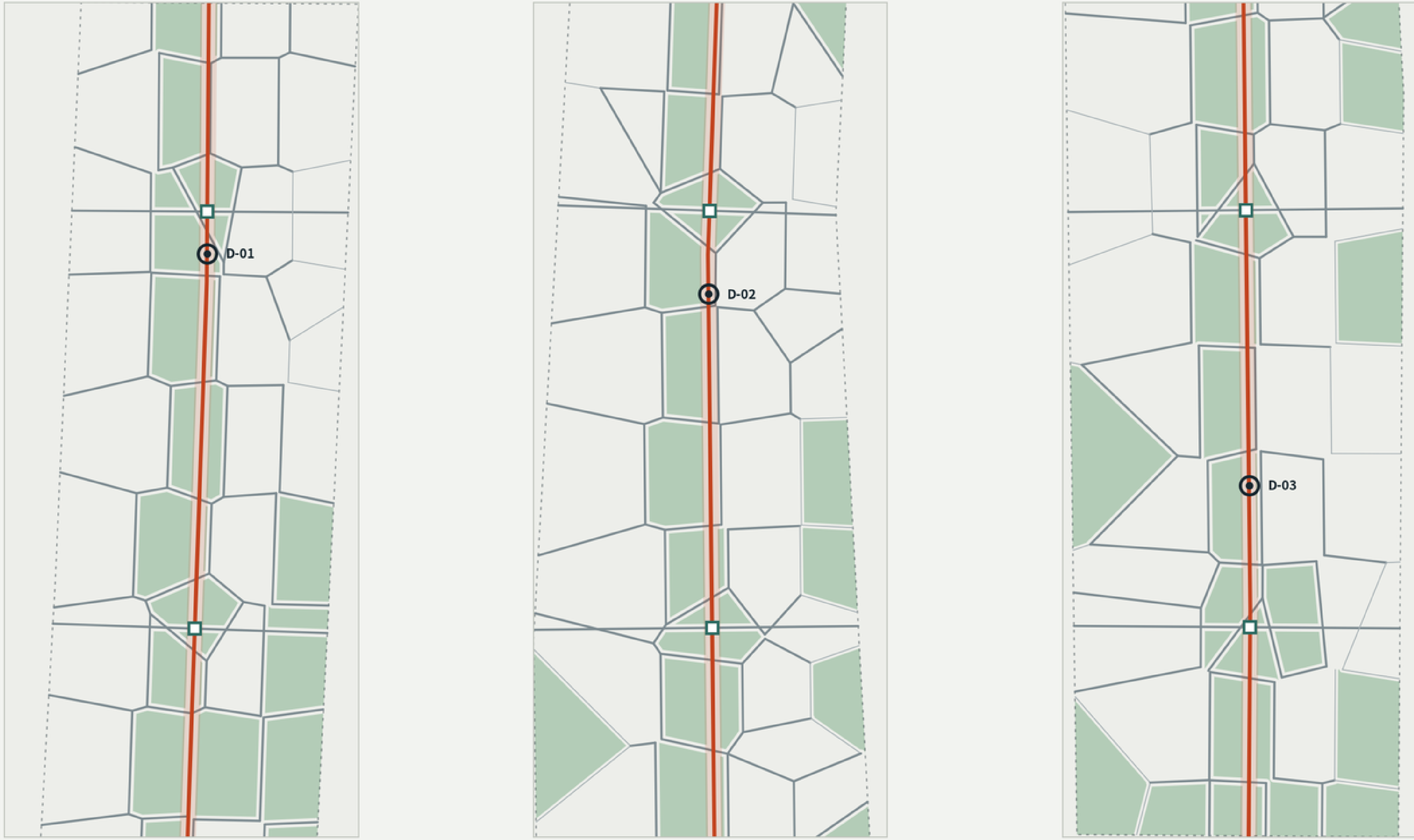
Fig. 4 Mobility and Blue-Green Public Space Structure

The road network follows parcel edges exactly; six stitch points are east-west crossings and physical demonstration sections of the component catalogue



Road area ratio 21.08% · Green ratio 26.16% · Public space ratio 4.59% · Right-of-way widths (44m / 26m / 16m) are conceptual assumed values pending official regulatory confirmation

Version Line (main slow-mobility spine) Secondary road Local street Stitch point Datum point Green space

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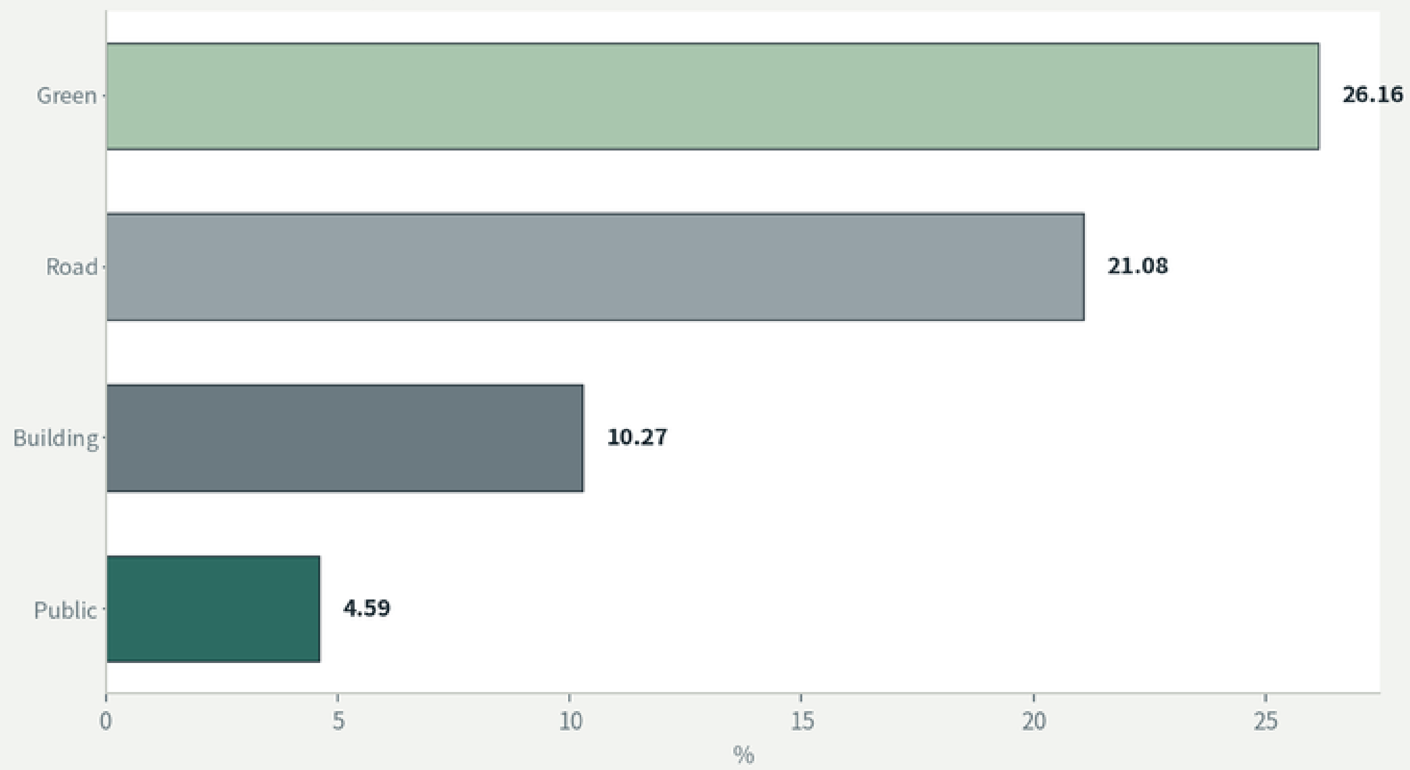
Fig. 5 Metric Recalculation and Evidence Chain

All metrics are recalculated from GeoJSON in EPSG:4548, never transcribed from narrative; missing official control indicators remain unknown and are not filled in

Evidence chain: from provisional boundary to verifiable metrics

Core ratio metrics (recalculated)



Metric status and confidence

Design area	**11.413 km²**	low	deviation 0.11%
Partition gap	**14.897 m²**	—	0.00013% of total; incomplete
Partition overlap	**0.099 m²**	—	0.00000087% of total; incomplete
Statutory FAR	**unknown**	—	official indicator missing; not filled in
Conceptual FAR	**0.923**	low	storey counts assumed
Spatial review	**PASS**	—	3 provisional notices only
Visual packaging	**PASS**	—	fully offline, no remote assets

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Component Catalogue and Datum System

agent.4 Public space, new business forms and pilgrimage landmarks

Catalogue as implementability

A component catalogue with specifications, price bands, maintenance liability and retirement clauses IS the delivery path — it can enter a government procurement process directly.

Decommissioning and rollback

The real risk of urban AI installations is not that they cannot be installed, but that once installed they cannot be removed, no one is accountable, and no one knows where the data went.

Writing down how to take something out demonstrates governance capability more than writing down how to put it in.

Spec card fields

Function definition and applicable scenarios
Physical specification (size, power, load, ingress protection)
Data behaviour (what is / is not collected; retention period)
Human-review triggers (when to hand over; who may disable)
Price band (conceptual range; not a quotation)
Whole life cycle (life, maintenance, responsible party)
Decommissioning and rollback (removal; data destruction)
Version and compatibility

Datum system

D-01 Datum Zero	physical release point of each version
D-02 The Objection Stand	physical registry for citizen objections
D-03 The Decommission Yard	displays retired parts and reasons
D-04 Contributor Stones	contributors inscribed by version

Showing failure is the source of this system's credibility.

AI Scenarios, Testing and Personas

agent.3 AI+ scenarios and a vibrant city

16 AI scenario cards

S-01 Lighting with visible version S-09 Accessibility guidance
S-02 Autonomous delivery shuttle S-10 Developer debug channel
S-03 Citizen fault reporting S-11 Objection intake
S-04 Layered heritage interpretation S-12 Decommissioning
S-05 Community service desk S-13 AI+Education assistant
S-06 Crowd guidance S-14 AI+Health triage support
S-07 AI-native commerce S-15 Robotics in public space
S-08 Waterfront monitoring S-16 Low-speed autonomous shuttle

Industrial test scenarios

T-01 Conformance testing lab certification results published
T-02 First-installation jury acceptability and real utility
T-03 Red-team stress testing published failure-mode list

No proposal dares state how its own system will break — which is exactly the dividing line of professionalism.

An active design constraint

No scenario card depends on individual identity recognition. This is an active design choice, not a passive compliance statement.

Eligibility, subsidy, dispute and disposition decisions always go to a human; critical public services keep a non-AI redundant path.

Personas

P-01 Specification engineer P-04 Student and developer
P-02 Hardware founder P-05 Procurement and maintenance
P-03 Proto Blocks resident P-06 International peer

Urban AI installations are ultimately procured and maintained by front-line units. Without P-05, implementability is empty.

Implementation, Operations and Data Gaps

agent.6 Global events and long-term operations

Phasing (conceptual)

Phase 1 Standard-setting conformance lab, certification and red-team capability, JZ-Parts v1.0 -> D-01 Datum Zero completed

Phase 2 First installation stitch-point sections, resident jury, scenario cards in batches -> D-02 Objection Stand opened

Phase 3 Volume deployment scale deployment of certified parts, AI-native business forms -> D-03 Decommission Yard opened

Long-term operations

Gauge Week: public version release + open conformance contest + retirement review
Developer community: open maintenance; public objection process
Scenario opening: Field Wing experimental zones, available on application, with mandatory publication of failure cases

Implementation policy

Certification replaces approval-based investment promotion:
Access Wing -> Gauge Works certification -> Proto Blocks -> Market Floor

No commitment is made on government investment, subsidies, tax preferences or the number of firms attracted.

Data gaps and recalculation

Official red lines and precise key-area polygons are not yet available. Five official control indicators - FAR, building height, density, green ratio and setback - all carry status missing; they remain unknown here and are not filled in.

Once official polygons are released, all GeoJSON layers, all area and ratio metrics, phasing, figures and the visualisation must be recalculated. Geometry generation is a parameterised script, so replacing the boundary source allows a single full re-run.